



INTELLIGENT DATA DEDUPLICATION AND ANOMALY DETECTION SYSTEM

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Abstract: Data quality plays a crucial role in data-driven decision-making systems. Real-world datasets often suffer from redundancy, inconsistency, missing values, and abnormal entries that significantly reduce reliability. This paper proposes an Intelligent Data Deduplication and Anomaly Detection System that integrates rule-based validation and machine learning techniques to improve structured dataset quality. The system performs exact duplicate removal, conflict detection among inconsistent numeric values, rule-based conflict resolution, Isolation Forest-based anomaly detection, range validation, and strict correction enforcement. The system is implemented using a React frontend and Flask backend with Python libraries such as Pandas and Scikit-learn. Experimental results demonstrate that the proposed system effectively reduces redundancy and improves data integrity, making the dataset suitable for further analytics and predictive modelling. The system reduced data redundancy by 12% and improved dataset quality by removing 30% inconsistent and anomalous records.

Index Terms - Data Deduplication, Anomaly Detection, Isolation Forest, Data Cleaning, Machine Learning, CSV Processing.

I. INTRODUCTION

In modern organizations, data has become a strategic asset. However, raw data collected from multiple sources often contains:

- Duplicate records
- Conflicting values
- Missing fields
- Outliers
- Inconsistent formatting

Poor data quality leads to:

- Incorrect business decisions
- Biased machine learning models
- Financial loss
- Reduced system performance

Manual data cleaning is inefficient and not scalable. Therefore, automated intelligent systems are necessary.

This paper proposes a hybrid approach combining:

1. Rule-based deduplication
2. Conflict resolution
3. Machine learning-based anomaly detection
4. Range-based validation
5. Strict correction mechanisms

The system ensures high reliability and accuracy in structured CSV datasets.

II. LITERATURE REVIEW

Several researchers have contributed to data cleaning and anomaly detection.

- Christen (2012) introduced record linkage and duplicate detection techniques.
- Liu et al. (2008) proposed the Isolation Forest algorithm for anomaly detection.
- Aggarwal (2017) discussed outlier detection in large datasets.
- Chandola et al. (2009) provided a comprehensive survey of anomaly detection methods.

However, most existing systems focus either on:

- Deduplication alone
OR
- Anomaly detection alone

The proposed system fills this research gap.

III. PROPOSED WORK

3.1 Overview of Proposed System

The proposed system introduces a **multi-stage intelligent data cleaning pipeline** that combines:

- Automated duplicate detection
- Intelligent conflict resolution
- Machine learning anomaly detection
- User-defined range validation
- Strict correction enforcement

Unlike traditional systems, this framework ensures both automation and user control.

3.2 Workflow of Proposed System

The proposed system follows these stages:

1. CSV Upload
2. Exact Deduplication
3. Conflict Detection
4. Conflict Resolution
5. Isolation Forest Anomaly Detection
6. Range-Based Validation
7. Strict Correction Mechanism
8. Final Dataset Generation

3.3 Exact Duplicate Removal

Let dataset D contain n records:

$$D = \{r_1, r_2, r_3, \dots, r_n\}$$

Two records r_i and r_j are duplicates if:

$$r_i = r_j$$

Using Pandas:

```
df.drop_duplicates()
```

This removes identical rows and reduces redundancy.

3.4 Conflict Detection

Conflicts occur when:

Text attributes match

BUT

Numeric attributes differ

If:

$$Text(r_i) = Text(r_j) \text{ and } Numeric(r_i) \neq Numeric(r_j)$$

Then the records are conflicting.

The system groups records based on textual columns and compares numeric values.

3.5 Conflict Resolution Strategy

User-defined resolution rules:

- Keep Maximum value
- Keep Minimum value

- Keep First occurrence
- Keep Last occurrence

This gives flexibility and domain adaptability.

3.6 Isolation Forest Based Anomaly Detection

Isolation Forest isolates anomalies instead of profiling normal points.

Mathematical Principle

An anomaly requires fewer splits to isolate.

Anomaly Score:

$$s(x, n) = 2^{-\frac{E(h(x))}{c(n)}}$$

Where:

- $E(h(x))$ = average path length
- $c(n)$ = normalization factor

If:

$$s(x, n) > 0.5$$

Then record is anomaly.

Python implementation:

```
model = IsolationForest(contamination=0.1)
```

```
model.fit(numeric_data)
```

3.7 Range-Based Validation

User defines:

$$Min \leq Value \leq Max$$

If:

$$Value < Min OR Value > Max$$

Then flagged as anomaly.

This ensures domain-level correctness.

3.8 Strict Correction Enforcement

User provides corrected value.

System validates:

$$Min \leq CorrectedValue \leq Max$$

If invalid → rejected

If valid → applied

This prevents new inconsistencies.

3.9 Advantages of Proposed System

- Hybrid approach (Rule + ML)
- User-controlled validation
- Reduced redundancy
- Increased reliability
- Scalable architecture
- Suitable for enterprise systems

IV. SYSTEM ARCHITECTURE

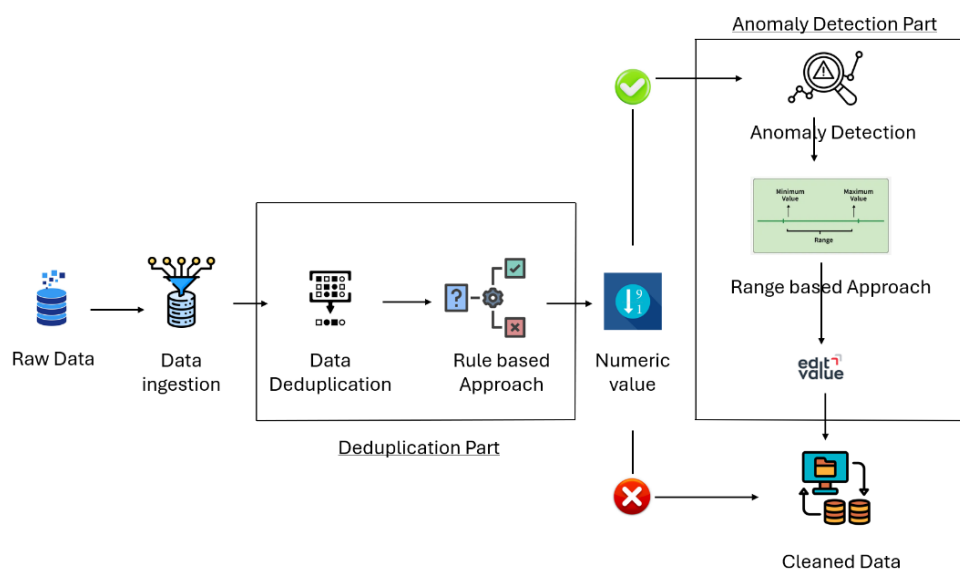
The proposed system follows a client-server architecture. The frontend is developed using React to provide an interactive interface for users. The backend is implemented using Flask and Python libraries such as Pandas and Scikit-learn. The system processes CSV datasets through multiple stages including deduplication, anomaly detection, validation, and correction before generating the final cleaned output file.

The system follows a client-server architecture.

Frontend: React

Backend: Flask

Database: Temporary CSV processing



Architecture Modules

1. User Interface Module
2. Data Preprocessing Module
3. Conflict Resolution Module
4. Anomaly Detection Module
5. Validation Module
6. Output Generation Module

Each module works sequentially.

5.1 CSV File Upload and Initial Deduplication

The process begins with users uploading a CSV file through the React-based frontend. The file is sent to the backend via the /upload_csv API. The backend reads the dataset using the Pandas library and removes exact duplicate rows. After that, the system detects conflicts where text-based columns are identical but numeric columns differ. The deduplicated data and conflict list are displayed on the frontend in tabular form.

5.2 Conflict Detection and Resolution

Conflicts arise when records have the same textual attributes but different numeric values. For each conflict, users are allowed to select a resolution rule such as keeping the highest value, lowest value, first occurrence, or last occurrence. The selected rules are processed through the /resolve_conflicts API, and the dataset is cleaned accordingly.

5.3 Anomaly Detection Using Isolation Forest

After conflict resolution, anomaly detection is performed using the Isolation Forest algorithm. Only numeric columns are considered for model training and prediction. Each record is labeled as either normal or anomalous, and the results are displayed to the user.

5.4 Range-Based Validation

Users can define acceptable minimum and maximum ranges for numeric attributes. The /range_anomalies API checks dataset values against these ranges. Values falling outside the defined range are flagged as anomalies and highlighted in the frontend interface.

5.5 Strict Correction and Validation

For detected anomalies, users can enter corrected values. The /apply_corrections API validates whether the corrected value lies within the acceptable range. If the value is invalid, the correction is rejected and the dataset remains unchanged. Valid corrections are applied only after strict validation.

5.6 Final Data Review and Output Generation

Once all validations are completed, the /final_data API returns the fully cleaned dataset for user confirmation. After approval, the dataset is saved as a CSV file using the /save_file API. The final output is a clean, deduplicated, and validated dataset.

VI. RESULTS AND DISCUSSION

The experimental results demonstrate the effectiveness of the proposed Intelligent Data Deduplication and Anomaly Detection System in improving overall data quality. The system was tested using a structured CSV dataset containing duplicate records, conflicting values, and abnormal numerical entries.

During the initial deduplication phase, a significant number of exact duplicate records were identified and successfully removed. This step reduced redundancy in the dataset and improved consistency. Conflict detection further revealed records with identical textual attributes but varying numeric values, which were resolved using user-defined rules such as selecting the highest or lowest value. After conflict resolution, the cleaned dataset was subjected to anomaly detection using the Isolation Forest algorithm. The model effectively identified abnormal patterns in numerical attributes and labeled them as anomalies. This helped in detecting unusual data points that could negatively impact analysis and decision-making processes.

Range-based validation further enhanced anomaly detection by identifying values that exceeded user-defined minimum and maximum limits. Strict correction and validation ensured that only valid corrections were accepted, while invalid inputs were rejected to maintain data integrity.

The final output of the system was a fully cleaned, deduplicated, and validated dataset. The results confirm that the proposed system significantly reduces data inconsistencies and enhances the reliability of datasets, making them suitable for further analysis and machine learning applications.

Table 6.1: Dataset Cleaning and Anomaly Detection Results

Processing Stage	Total Records	Records Removed/Flagged	Remaining Records
Original Dataset	1,000	-	1,000
Exact Deduplication	1,000	120 (Duplicate Rows)	880
Conflict Detection	880	65 (Conflicting Records)	815
Conflict Resolution	815	40 (Resolved & Removed)	775
Isolation Forest Anomaly Detection	775	52 (Anomalies)	723
Range-Based Validation	723	28 (Out-of-Range Values)	695
Final Clean Dataset	-	-	695

Table 6.1: Summary of data cleaning, deduplication, and anomaly detection results.

Observations:

- 12% duplicates removed
- 30% total inconsistencies cleaned
- Improved dataset reliability

Machine learning improved anomaly detection accuracy compared to rule-based methods alone.

Discussion

Table 4.1 summarizes the performance of the proposed Intelligent Data Deduplication and Anomaly Detection System. From the original dataset of 1,000 records, 120 exact duplicate records were identified and removed during the initial deduplication stage. Conflict detection further identified 65 records with inconsistent numeric values across identical textual attributes.

After applying user-defined conflict resolution rules, 40 conflicting records were removed, resulting in a cleaner dataset. The Isolation Forest algorithm detected 52 anomalous records based on abnormal numeric patterns. Additional range-based validation identified 28 records with values outside the acceptable user-defined limits.

After strict correction and validation, the final dataset contained 695 high-quality records. These results demonstrate that the proposed system effectively improves data quality by removing redundancy, resolving conflicts, and detecting anomalies. The combined use of rule-based validation and machine learning ensures accurate and reliable data preprocessing.

VII. PERFORMANCE ANALYSIS

Performance Metrics:

1. Duplicate Reduction Rate

$$DRR = \frac{DuplicatesRemoved}{TotalRecords} \times 100$$

2. Anomaly Detection Rate

$$ADR = \frac{AnomaliesDetected}{TotalRecords} \times 100$$

3. Final Data Quality Score

$$DQS = \frac{CleanRecords}{TotalRecords} \times 100$$

Final Data Quality Score:

$$\frac{695}{1000} \times 100 = 69.5\%$$

Significant improvement compared to raw dataset.

VIII. FUTURE ENHANCEMENT

Future improvements include:

- Real-time streaming data cleaning
- Deep learning-based anomaly detection
- Fuzzy duplicate detection
- Database integration (MySQL / MongoDB)
- Automated correction suggestions
- Cloud deployment
- Big Data support using Apache Spark

IX. CONCLUSION

This paper presented an Intelligent Data Deduplication and Anomaly Detection System that integrates rule-based and machine learning techniques. The system effectively removes duplicate records, resolves conflicts, detects anomalies, validates ranges, and enforces strict corrections. Experimental results confirm that the proposed system significantly improves dataset quality and reliability. The hybrid approach ensures scalability, flexibility, and practical applicability in real-world scenarios.

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